STAGE 2 – PRACTICAL ENGINEERING CHALLENGE

TEAM ID:	TEAM NAME:
LRC-26-0047	TEAM DISAPPOINTMENT

Problem Statement: Industrial Manufacturing Process Automations

Industrial manufacturing depends on highly controlled liquid-handling systems to deliver process fluids reliably from a source or bulk supply to downstream process equipment. Depending on the application, these fluids may include **ultrapure fluids, process chemicals, solvents, developers, or CMP-related fluids**. Industry standards explicitly address liquid-chemical distribution, high-purity piping, bulk chemical distribution, system qualification, and UPW system operation and monitoring.

In such systems, **flow is a fundamental process variable.** An incorrect flow rate can indicate or cause a restriction, incorrect valve state, inadequate pumping performance, abnormal process demand, or another hydraulic deviation. The consequence may be unstable process conditions, incorrect chemical delivery, equipment malfunction, or loss of process performance.

This challenge therefore focuses on a fundamental industrial facilities and process-engineering task: **To establish a defined process-fluid flow, measure it reliably, detect a deviation, diagnose its cause, and restore the process to the required operating condition.**

Engineering Challenge

As you are part of a multidisciplinary engineering team responsible for designing, building and validating a small-scale automated liquid-flow system represented of industrial process handling system.

CASE 1 – CONTROL + MEASUREMNT

Problem: Develop a system that manages the changing level of fluid in the sump automatically (when there is a sudden external fluid charging to the sump) and while accurately measuring and totalizing the fluid discharged from the sump using the flow transmitter.

CASE 2 – FLUSHING +SHUTDOWN

Problem: Develop an alternative operating configuration that allows flush water to pass through the system while all three pumps operate simultaneously, and then safely transition the system to an OFF condition when the flush-water supply has been exhausted.

Given Process Parameters

Pipe diameter =	8mm
Fluid density (ρ) =	1000kg/m^3
Dynamic Viscosity (μ) =	1x10^{-3}Pa.s
Required Flow range =	1-6 L/min
Nominal target flow for the live challenge =	3L/min

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Available Hardware

Mechanical Hardware	Electronic Hardware
1. Water Tank – 3 Quantity 2. Pumps – 3 Quantity 3. FT-01 Flow Transmitter – 01 Quantity 4. Ball valve – 3 Quantity 5. ON/OFF/XV Valve – 1 Quantity 6. 10-mm OD tubing/blue – 9 meters/1 meter 7. Microcontroller/PCB/ data-acquisition software 8. Required electrical/ electronic accessories	1. Custom PCB 2. ESP32 NodeMCU module 3. Pump Driver-1 4. Pump Driver-2 5. Pump Driver-3 6. 12 V DC power input (**SMPS**) 7. 12 VDC Solenoid-valve interface 8. Flow-sensor interface 9. Float-switch interface 10. ON/OFF-switch interface 11. Pump-driver PWM/enable interfaces 12. On/Off Switch

The team must determine the appropriate arrangement, connections and measurement method.

- Assemble and commission the liquid-flow system using the components provided.
- Measure and characterize the flow using FT-01
- Interface FT-01 with the provided **data-acquisition system**
- Experimentally determine the relationship between sensor output and actual flow.
- Characterize the effect of valve position on flow rate. Record and analyse flow vs time data
- Maintain target flow of 3L/min
- Use graphs and measured data to justify engineering decisions
- Design an appropriate hydraulic arrangement and P&ID

TEAM INSTRUCTIONS

- As a team go through the list provided to read about each component that has been Listed.
- The PCB is a 12 V DC and the SMPS connections should be made carefully.

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TASK 1: HYDRAULICS SYSTEM DESIGN

Q1.1 Design and build the hydraulic path with the flow transmitter on the recirculation line. Submit/sketch a labelled P&ID below by showing the pumps, Needle valves, flow transmitter, flow direction, Tank, sump, tube diameter and measurement points.

CASE 1 – Process Fluid Transfer & Automatic Level Protection

Objective: Set-up and operate a closed-loop process- fluid handling system using the available tank, pump, valves flow transmitter, sump, float sensor and control PCB. The System must continuously transfer process fluid recirculation system by connecting the 10L fluid tank with a On/Off valve to a pump-01 and the pump-01 goes to the inlet of sump 4.2L tank. The Sump 4.2L outlet line is connected to another pump-02 followed by a flow transmitter that measures the total flow back to the 10L fluid tank. The Pump-01 runs on 50% duty cycle and Pump-02 runs on 100% duty cycle. **Calibrate the flow transmitter** and demonstrate correct fluid circulation, if there is sudden introduction of external fluid into the Sump tank then Pump-01 should stop while pump – 02 should still runs till fluid level reduces in the sump. After the fluid reduction in the SUMP 4.2L tank the Pump-01 should turn on and the process must continue. **Also, demonstrate Flow vs Time trends in mL/s, total transferred Liters back to the 10L tank** (share Screenshots flow in L/min). **Share the code**. The case-1 can be turned on using the available on/off Switch, to start the system.

CASE 2 – Flush/Line Conditioning Sequence

Objective: After successfully completing case 1, initiate the flush water in the sequence using the **equal Tee.** The On/Off valve must be switched off in order to isolate the 10L process fluid tank. The flush Fluid tank (4.2L) is used as a temporary fluid source to flush out the process fluid or any debris from the system. Before the start of this process make sure you open the manual ball valve on the fluid flush line before Pump-03 starts. The entire process of case -1 and case - 2 should run in a single go without manual intervention except opening of ball valve before the Pump-03 starts. **The System must stop as soon as the tank flush fluid 4.2L tank finishes.** The system must be automated without human interventions (only ball valve can be opened manually before the start of flush line sequence)

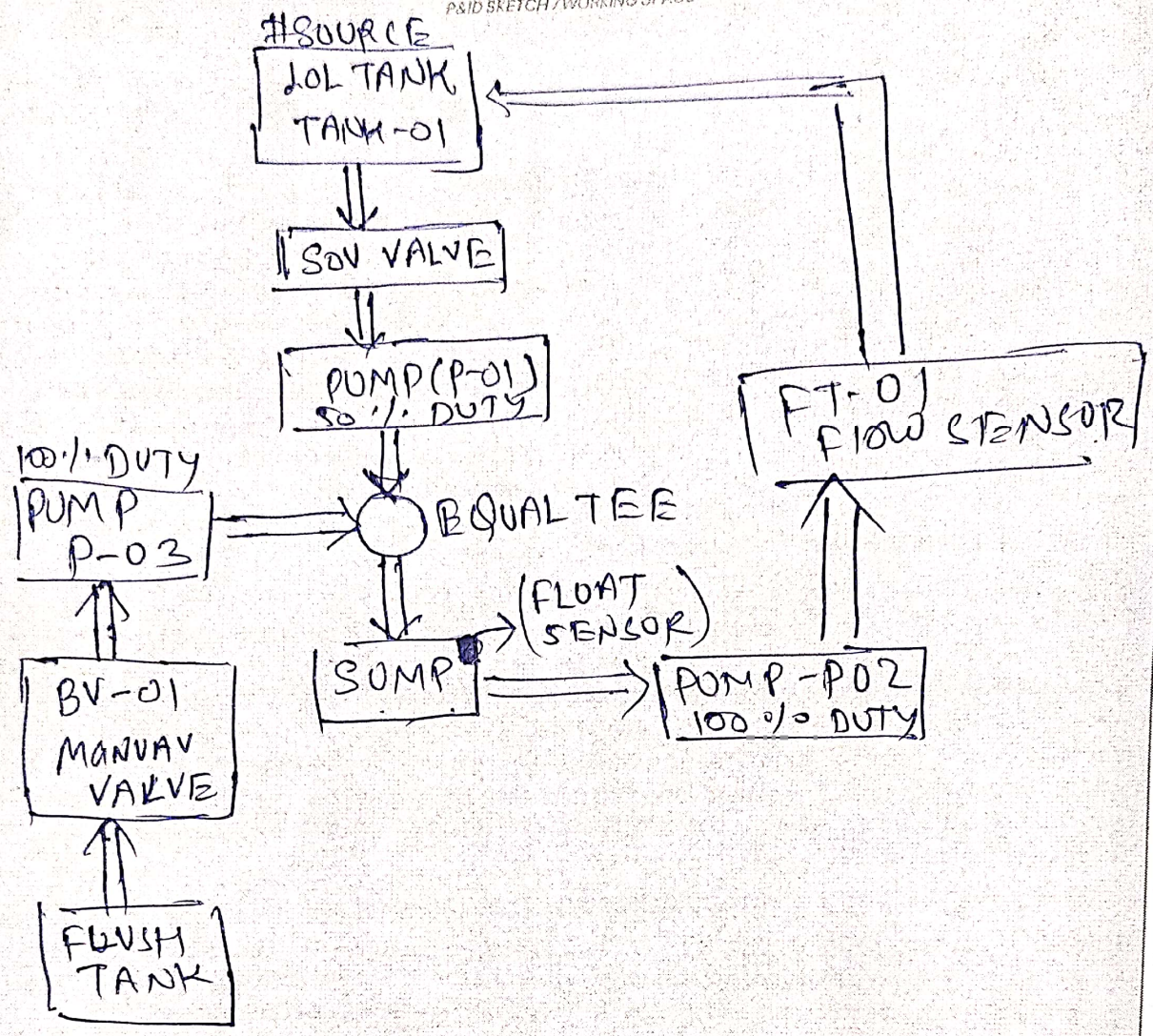
SUBMIT:

A. Submit: 10-15 min Video and Photograph of the completed setup, with identification of each component and explain the direction of flow in the video verbally. Demonstrate both Case - 1 and Case – 2.

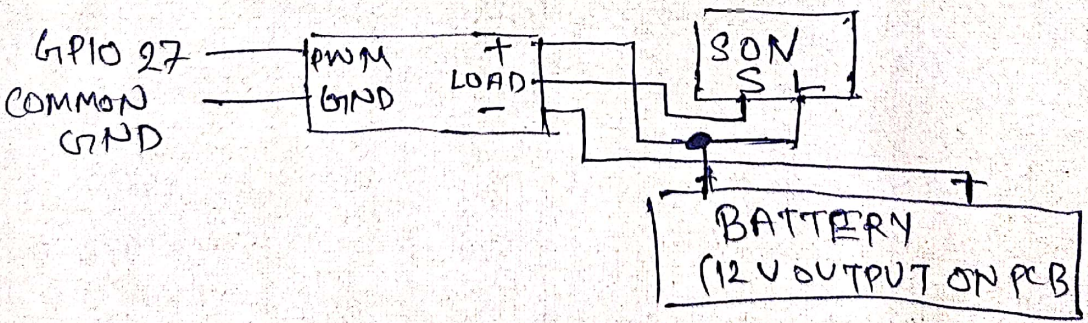
B. Final P&ID Block diagram

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P&ID SKETCH / WORKING SPACE



CONNECTIONS FOR SOV + MOSFET DRIVER



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Q1.2 After setting-up the system, now operate it at approximately at 1L/min, 3L/min & 6L/min. Compare the theoretical velocity with the actual operating behaviour.

WORKING / COMPARISON

At 1 L/min: the pump was operated at ~~20% PWM~~, ~~Producing a m~~ 36% PWM with pulse frequency of 110 producing roughly flow rate (Q = 1.052 L/min) A PID controller could be implemented to automatically adjust the speed (duty cycle) based on the reference (ref) and (output).

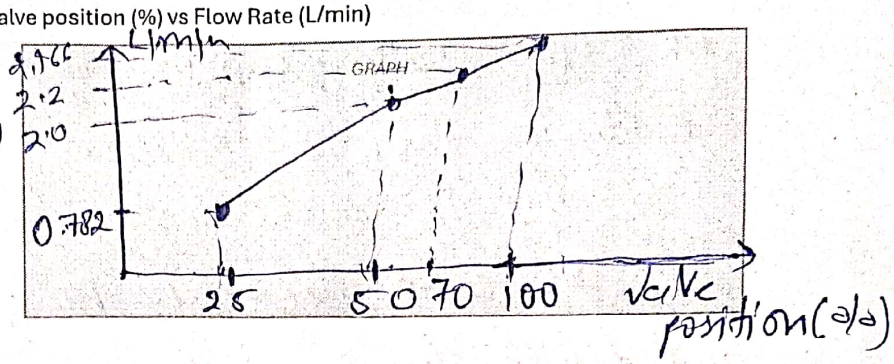
At 3 L/min & 6 L/min: The pump was when operated at 100% PWM, producing a maximum measured flow rate of approximately 2.438 L/min. the target of 3 or 6 L/min was not achievable with 12 V output. As the rated voltage was max = 14.4 V, running at at max may have given us the desired results.

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FLOW RESPONSE TO VALVE POSITION

Q 2.1 Set the ball valve to approximately 25%,50%,75% & 100%. At each position, allow the flow to stabilize and record the flow rate. Create the following graph of valve position (%) vs Flow Rate (L/min)

VALVE POSITION	FLOW RATE
25%	0.782 L/min
50%	2.0 L/min
70%	2.2 L/min
100%	2.466 L/min



Q2.2 Determine the experimental calibration equation: $Q = af + b$, where Q = flow rate in L/min, f = pulse frequency in Hz, a = experimental calibration coefficient and b = offset.

Report:

R^2 =	0.9972
a =	0.008176
b =	0.151951
Maximum calibration error =	0.0660 L/min
Average calibration error =	0.0278 L/min

Q2.3 FT-01 produces pluses corresponding to water flow. Run the system at least for 5 different flow conditions within the available operating range (Hint: use formulas f=N/t and Q=V/t). For each condition, record the following:

Trial	Pulses	Time(s)	Frequency (Hz)	Collected Volume (L)	Actual Flow (L/min)
1	3664	~~80.50~~ 85.22	43	0.755	0.532
2	5047	40.70	124	0.746	1.100
3	6439	34.62	186	0.986	1.709
4	7619	32.29	236	1.123	2.087
5	9160	32.71	280	1.329	2.438

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CALCULATIONS / OBSERVATIONS

2.1: Calculations → Flow rate was estimated by the caliberation equation 'q = 0.008176 f + 0.152296', for 25%, 50%, 70%, 100%, f = 77, 226, 251, 283 respectively.

Observation → Flow rate increased with valve opening although rate is not uniform.

2.2: Calculations: The 5 experimental readings from '2.3' were used for linear regression to obtain the caliberation equation: $q = af + b$,

a = slope, b = intercept of the fitted line, R^2 = 0.9972, indicates how well the calliberation line fits the experimental data.

Observation: High R^2 indicates a strong linear relationship between pulse freqency and flow rate. The maximum & average caliberation errors, which were calculated by comparing the flow rates predicted by the equation with experimentally measured flow rates, are less which shows the calliberated equation predicts the flow rate with good accuracy.

2.3: observation: The maximum measured flow was below the target of 3L/min or rated value of 6L/min

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TASK 1: HYDRAULIC SYSTEM THEORY AND FAULT ANALYSIS

Q3.1 The effective liquid volume inside a section of tubing is 0.5L, while the operating flow rate is 2.5L/min. The hydraulic residence time is given by $T_r = Q/V$. Calculate the residence time in seconds, also answer the following questions:

1. What does residence time physically represent in this system?
2. What happens to residence time if the tubing volume is doubled while the flow rate remains constant?
3. What happens to residence time if the flow rate is doubled while the tubing volume remains constant?
4. Why is residence time important when interpreting a sudden change in measured flow?

ANSWER Given $V = 0.5\,l$, $Q = 2.5\,l/min$ $T_r = V/Q = 0.5/2.5$

$T_r = 12$ seconds $= 0.20$ mins

(1) It is average time needed to completely replace the fluid liquid volume inside the tubing section at the given rate essentially how long it takes for a 0.5L parcel of fluid to sweep through and exit. It is a nominal average actual individual parcel times vary slightly with velocity profile and mixing

(2) $T_{r,new} = 2V/Q = 2 \times 12.5 = 24.5$ twice the volume at same rate

(3) $T_{r,new} = V/2Q = 12/2 = 6$ sec higher throughput sweeps the same volume twice ~~the~~ as fast

(4) T_r is the transport lag for a bulk fluid change (concentration front, temperature step) entering one pipe length upstream it takes 12 s to reach FT-01. FT01 it itself responds to local pressure changes almost instantly, so $T_r \neq$ sensor response delay. confusing the two leads to wrong fault onset timestamping. In our runs, adjusting the ball valve caused a brief settling period before FT-01 stabilised partly transport lag, partly ESP32 firmware signal filtering

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Q3.2 The following observation are obtained from the same system:

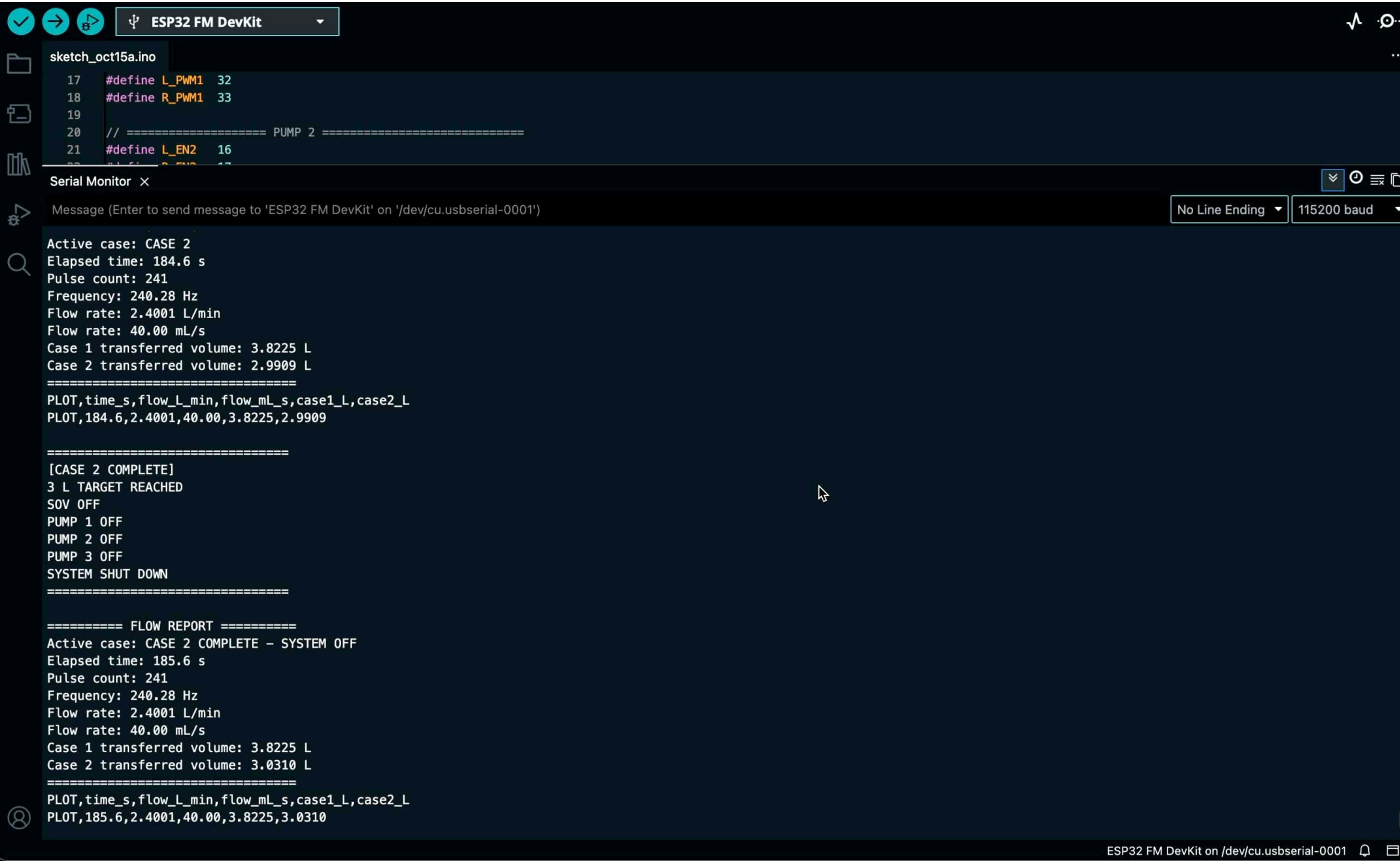
CONDITION	PUMP	VALVE	FT-01 PULSE SIGNAL	MEASURED FLOW
A	ON	Open	Normal	Normal
B	ON	Partially closed	Reduced	Reduced
C	ON	Open	Zero	Zero
D	OFF	Open	Zero	Zero
E	ON	Open	Intermittent	Fluctuating

For each of the condition, identify the most likely physical or instrumentation condition. Then answer: Which condition is most difficult to distinguish using FT-01 alone, and what additional information would you require to distinguish the fault? (Hint: Teams must consider possibilities such as: Valve restriction, pump failure, hydraulic blockage, sensor/signal failure, unstable flow etc)

ANSWER A: Normal condition, everything working, no action.
B: Reduced Pulse signal & measured flow means valve restriction causing reduced flow.
C: Pump failure despite the ON command, sensor wiring fault, ESP32 pin conflict (wrong pin) or not working, physical blockage/hydraulic blockage
D: Zero flow rate with pump OFF is normal, no action required.
E: Unstable flow/signal of flow sensor problem, air slugs in tubing
Hardest to distinguish: - Condition 'C': - Blockage, pump failure, wiring fault & firmware bug all produces identical zero output - FT01 alone can't separate.
To distinguish: i) Collect outlet fluid: flow out ⇒ sensor fault, nothing out ⇒ blockage or pump failure
ii) Pressure both sides of suspected blockage: high upstream + zero upstream ⇒ blockage

iii) Raw GPIO on ESP32: read flow sensor directly via pin — pulses present but uncounted ⇒ firmware fault, flat line ⇒ wiring issue iv) Voltage across sensor (may get damaged

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